

MSTAR PRACTICE WORKSHEET**Lesson 3-2: Understand Rates and Unit Rates**

Grade 6 Mathematics · Standard 6.AT.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

A booth charges \$4.50 for 9 games. What is the unit rate?

- ☐ A \$0.45 per game
- ☒ B \$0.50 per game
- ☐ C \$2.00 per game
- ☐ D \$4.50 per game

$\$4.50 \div 9 = \0.50 per game.

Question 2 **SELECTED RESPONSE — CORRECT: B**

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

- ☐ A 6 miles per gallon
- ☒ B 30 miles per gallon
- ☐ C 36 miles per gallon
- ☐ D 180 miles per gallon

$180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon.}$

Question 3 SELECTED RESPONSE — CORRECT: B

A pack of 8 markers costs \$6.00. What is the cost per marker?

- ☐ A \$0.60
- ☒ B \$0.75
- ☐ C \$0.80
- ☐ D \$1.33

$\$6.00 \div 8 = \0.75 per marker.

Question 4 SELECTED RESPONSE — CORRECT: A

Which is a unit rate?

- ☒ A 12 miles per 1 hour
- ☐ B 24 miles in 2 hours
- ☐ C 36 miles for 3 gallons
- ☐ D 100 points in 5 games

A unit rate has 1 in the second quantity. '12 miles per 1 hour' is the only choice where the second quantity is 1.

Question 5 SELECTED RESPONSE — CORRECT: A

Token packs: Pack A is 20 tokens for \$5.00, Pack B is 35 tokens for \$7.00, and Pack C is 50 tokens for \$12.50. Which pack is the best deal per token?

- ☒ A Pack B
- ☐ B Pack A
- ☐ C Pack C
- ☐ D They are all the same

Pack B has a unit rate of \$0.20 per token, which is the lowest cost per token.

Question 6

SELECTED RESPONSE — CORRECT: A

Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

- ☒ A Runner B
- ☐ B Runner A
- ☐ C Same rate
- ☐ D Cannot determine

Runner A: $4 \div 12 \approx 0.33$ laps/min. Runner B: $5 \div 14 \approx 0.36$ laps/min. Runner B runs slightly more laps per minute.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: B

Booth A charges \$3 for 5 games. What is the unit rate in dollars per game?

- ☐ A \$0.30 per game
- ☒ B \$0.60 per game
- ☐ C \$1.67 per game
- ☐ D \$15.00 per game

Unit rate = total cost $\div$ number of games = $3 \div 5 = \$0.60$ per game.

- **A:** That divides 3 by 10. There are 5 games, so divide 3 by 5.
- **C:** That is 5 divided by 3, the ratio upside down. Dollars per game divides dollars BY games.
- **D:** That multiplies 3 by 5. A unit rate divides the total cost by the number of games.

Part B — correct answer: D

Booth B charges \$5 for 8 games. Which booth is the better deal, and why?

- ☐ A Booth B, because \$5 buys more games than \$3.
- ☐ B Booth A, because 3 is less than 5.
- ☐ C They are equal, because both ratios simplify to the same value.
- ☒ D Booth A, because \$0.60 per game is less than \$0.625 per game.

Booth B's unit rate is $5 \div 8 = \$0.625$ per game. Comparing per-game costs, $\$0.60 < \0.625 , so Booth A costs less per game.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Dividing the Wrong Way

Marisol wanted the cost per pound of apples. The sign read '4 pounds for \$6.' She wrote: 'Unit rate = $4 \div 6$ = about \$0.67 per pound.'

Explain Marisol's misconception and give the correct unit rate.

Model answer: $6 \div 4 = \$1.50$ per pound. The words 'per pound' say what to divide by: dollars $\div$ pounds. Marisol's $4 \div 6 \approx 0.67$ is pounds per dollar, a different rate that answers a different question.

2 points	Student identifies that 'cost per pound' means dollars divided by pounds, computes $6 \div 4 = \$1.50$ per pound, and explains that $4 \div 6$ gives pounds per dollar instead.
1 point	Student gives \$1.50 per pound without explaining which quantity is divided by which.
0 points	Student agrees with Marisol or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Model answer: The student is not correct. Dividing each price by its tokens gives $\$5 \div 25 = \0.20 per token, $\$10.80 \div 60 = \0.18 per token, and $\$19 \div 100 = \0.19 per token. The 60-token pack has the lowest unit rate, so it is the best deal — getting the most tokens is not the same as paying the least for each one.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.